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Faculty of Science and Technology

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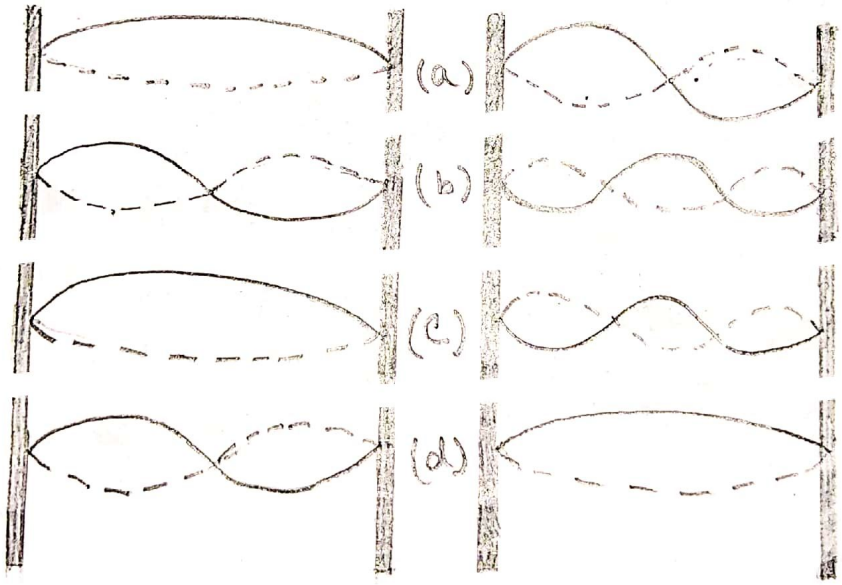
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Problem 1: Strings A and B have identical and linear densities, But string is under greater tension than string A. Figure shows four situations (a) through (d), in which standing wave patterns exists on the two strings. In which situations is there the possibility that string A and B are oscillating at the same resonant frequency?



Answer: We know, $v = \sqrt{\frac{T}{\mu}}$, B has greater T, so v is greater in B. But v also is equal to λf ($v = \lambda f$) or $f = \frac{v}{\lambda}$. So to get the same frequency when v is greater λ must be shorter. Only situation to be is (d).

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Problem 2: A generator at one end of a very long string creates a wave given by

$$y = (6.0\ cm)\cos\frac{\pi}{2}\left[(2.00\ m^{-1})x + (8.00\ s^{-1})t\right]$$

and a generator at the other end creates the wave,

$$y = (6.0\ cm)\cos\frac{\pi}{2}\left[(2.00\ m^{-1})x - (8.00\ s^{-1})t\right]$$

calculate the (a) frequency, (b) wavelength (c) speed of each wave. For $x \geq 0$, what is the location of the node having the (d) smallest, (e) second smallest and (f) third smallest value of x? For $x \geq 0$, what is the location of the antinode having the (g) smallest, (h) second smallest, (i) third smallest value of x?

Answer: From the given equation the wave number is, $K = \frac{2\pi}{\lambda}$

$$= \pi\ rad/m$$

Angular frequency,

$$\omega = 2\pi f = 4\pi\ rad/s$$

Given,

$$y = (6.00\ cm)\cos\frac{\pi}{2}\left[(2.00\ m^{-1}) + (8.00\ s^{-1})t\right]$$ and

$$y = (6.00\ cm)\cos\frac{\pi}{2}\left[(2.00\ m^{-1})x - (8.00\ s^{-1})t\right]$$

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(a) The wave frequency $f = \frac{4\pi}{2\pi}$

$= 2$ Hz (Ans:)

(b) The wavelength, $\lambda = \frac{2\pi}{\pi}$

$= 2m$ (Ans:)

(c) The speed of the wave is,

$$v = \frac{\omega}{k} = \frac{4\pi}{\pi} = 4 m/s$$

(Ans:)

(d) For any node on the string at any instant of time $y = 0$; By substituting at $t = 0$ and $y = 0$, we have $0 = \cos(\pi x)$

Solving for x, $\pi x = n\frac{\pi}{2}$; where $n = 1, 3, 5$

Therefore $x = \frac{n}{2}$

The smallest x at which the node exist is at $\boxed{n = 1 \text{ at } x = 0.5 \text{ m}}$ (Ans:)

(e) The second smallest x at which a node exist is at $\boxed{n = 3 \text{ at } x = 1.5 \text{ m}}$

(Ans:)

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(f) The third smallest x at which a node exist is at $\boxed{n=5 \text{ at } x=2.5\,m}$ (Ans:)

(g) For any anti-node on the string we have $y = \pm y_m$; By substituting at $y = \pm 6$ cm

we have $\pm 1 = \cos(\pi x)$

Solving for x $\pi x = n\pi$, where $n = 0, 1, 2, 3$

Therefore $x = n$

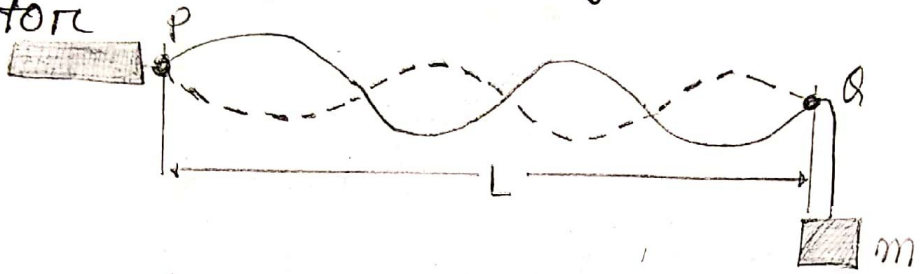
The smallest value of x that corresponds to an anti-node (maximum) is at $n = 0$ where $x = 0\,m$ (Ans:)

(h) The second smallest value of x that corresponds to an anti-node is at $n = 1$ where $x = 1\,m$ (Ans:)

(i) The third smallest value of x that corresponds to an anti-node (maximum) is at $n = 2$, where $x = 2\,m$.

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Problem 3: In Fig. a string, tied to a sinusoidal oscillator at P and running over a support at Q, is stretched by a block of mass m. Separation L = 1.20 m, linear density $\mu = 1.6\ g\,m^{-1}$ and the oscillator



frequency f = 120 Hz. The amplitude of the motion at P is small enough for that point to be considered a node. A node also exist at Q. (a) What mass allows the oscillator to set up the fourth harmonic on the string? (b) What standing wave mode, if any, can be set up if m = 1.00 kg?

Answers: We Know,

(a) $$f = \frac{nv}{2L} = \frac{n}{2L}\sqrt{\frac{\tau}{\mu}}$$

$$\Rightarrow f = \frac{n}{2L}\sqrt{\frac{mg}{\mu}}$$

$$\Rightarrow m = \frac{4L^2 f^2 \mu}{n^2 g}$$

$$\Rightarrow m = \frac{4 \times (1.2)^2 \times (120)^2 \times 1.6}{(4)^2 \times 9.8}$$

Given,

$L = 1.2\ m$

$\mu = 1.6\ g/m$

$f = 120\ Hz$

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$\therefore m = 0.846 \text{ kg}$

$\approx 0.85 \text{ kg}$ (Ans:)

(b) We know,

$$f = \frac{n}{2L}\sqrt{\frac{mg}{\mu}}$$

$$\Rightarrow n = \sqrt{\frac{4L^2 f^2 \mu}{mg}}$$

$$= \sqrt{\frac{4\times(1.2)^2\times(120)^2\times 1.6\times 10^{-3}}{1\times 9.8}}$$

$= 3.68$ is an integer

$\therefore$ There will be no mode to be set up by oscillator because the result is against the claim.

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Problem 4: An automobile can be mounted on four identical as far as vertical oscillations are concerned. The spring of a certain car are adjusted so that the oscillations have a frequency of 3.00 Hz. (a) What is the spring constant of each spring if the mass of the car is 145 kg and the mass is evenly distributed over the springs? (b) what will be the oscillation frequency if five passengers, averaging 73.0 kg each, ride in the car with an even distribution of mass?

Answer: (a) Let K be the spring constant for each of the springs. Force acting on the car is $F = -4Kx$ and angular frequency is $\omega = \sqrt{k/m}$ ∴ For this car,

$$\omega = \sqrt{\frac{4K}{m_{car}}}$$

Solving for K and substitute with $\omega = 2\pi f$ ∴ we get,

$$4\pi^2 f^2 = \frac{4K}{m_{car}}$$

$$\Rightarrow K = \pi^2 f^2 m_{car}$$

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Substitute with the givens to get,

$$K = \pi^2 (3.00\ Hz)^2 \times 1450\ kg$$

$$\therefore K = 1.29 \times 10^5\ N/m \quad \text{(Ans:)}$$

(b) Now if five passengers averaging 73.0 kg each, ride in the car with an even distribution of mass then the new mass on the plateform will be, $m_{new} = ?$

$$\therefore m_{new} = 362.5 + \frac{5}{4}(73)$$

$$= 453.75\ kg$$

$m = 1450\ kg$

$\frac{1}{4} m = 362.5\ kg$

$$\therefore f = \frac{\omega}{2\pi}$$

$$= \frac{16.86}{2\pi}$$

$$= 2.68\ sec \quad \text{(Ans:)}$$

$$\omega = \sqrt{K/m}$$

$$= \sqrt{1.29 \times 10^5 / 453.75}$$

$$= 16.86\ rad/s$$

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Problem 5: A block rides on a piston (a squat cylindrical piece) that moving vertically with simple harmonic motion. (a) If the SHM has period 1.0 second, at what amplitude of motion will the block and piston separate? (b) If the piston has an amplitude of 5.00 cm, what is the maximum frequency for which the block and piston will be in contact simultinously?

Answer: (a) The block and piston will separate if the acceleration of the piston greater than the acceleration of gravity. So the acceleration of gravity equals piston's acceleration, that is,

$$\omega^2 x_m = g$$

In terms of the period, the angular frequency can be written as, $\omega = \frac{2\pi}{T}$

$$\text{So,} \quad \Rightarrow \left(\frac{2\pi}{T}\right)^2 x_m = g$$

Solving for x_m then substitute with the givens,

$$x_m = \frac{gT^2}{4\pi^2} = \frac{(9.8)\times(1)}{4\pi^2}$$

$$\therefore x_m = 0.25 \text{ m} \quad \text{(Ans:)}$$

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(b) Now if the piston has an amplitude of $x_m = 5.0\ cm$, we need to find the maximum frequency such that the block and piston will be in contact continuously, to find the frequency we set the acceleration of gravity equals to the piston acceleration and solve for f.

Here

$$\omega^2 x_m = g$$

$$\Rightarrow (2\pi f)^2 x_m = g \qquad \text{here } \omega = 2\pi f$$

$$\Rightarrow 2\pi f = \sqrt{\frac{g}{x_m}}$$

$$\Rightarrow f = \frac{1}{2\pi}\sqrt{g/x_m}$$

$$\Rightarrow f = \frac{1}{2\pi}\sqrt{9.8/0.05m}$$

$$\therefore f = 2.23\ Hz$$

(Ans:)

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Problem 6 : Figure 15-38 gives the one dimensional potential energy well for a 2.0 kg particle (the function $U(x)$ has the form bx^2 and the vertical axis scale is set by $U_s = 2.0\,J$). (a) If the particle passes through the equilibrium position with a velocity of 85 cm/s, will it be turned back before it reaches $x = 15$ cm? (b) If yes, at what position. If no, what is the speed of the particle at $x = 15$ cm?

Answer: Given,

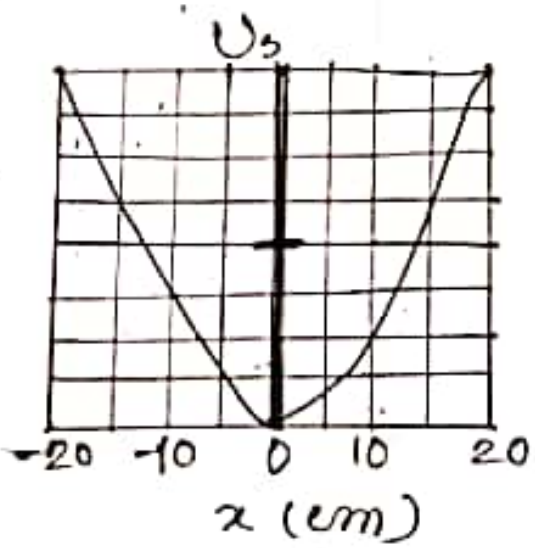
$U_s = 2.0\,J$

$x = 20$ cm

$U(x) = bx^2$

$\therefore 2.0 = b(0.2)^2$

$\Rightarrow b = 50.$



(a) We know,

$E = U + K$

Now, $K = \frac{1}{2}mv^2 = \frac{1}{2} \times (2) \times (0.85)^2$

$= 0.7225\,J$

$m = 2$ kg

$v = 85$ cm/s

$= 0.85$ m/s

$\therefore U = bx^2 = 50 \times (0.15)^2 = 1.125\,J$ [Yes! it will]

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(b) Now we have to set the potential energy equal to 0.7225

$$\therefore 0.7225 = 50x^2 \quad [b = 50]$$

$$\Rightarrow x^2 = \frac{0.7225}{50}$$

$$\Rightarrow x^2 = 0.01445$$

$$\Rightarrow x = 0.12 \text{ m}$$

$$\therefore x = 12.0 \text{ cm}$$

(Ans:)

Problem 7: At a certain harbor, the tides cause the ocean surface to rise and a distance d (from highest level to lowest level) in simple harmonic motion, with a period of 12.5 h. How long does it take for the water to fall a distance 0.250 d from its highest level?

Answer: It is known that highest level to the lowest level equals twice the amplitude. So the amplitude of the wave

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is $x_m = \frac{1}{2}d$ and the angular frequency is

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{12.5h} = 0.503 \text{ rad/h}$$

The coordinate of the wave is,

$$x = x_m \cos(\omega t + \phi) = \frac{1}{2} d \cos(\omega t + \phi)$$

at $t = 0$; when $x = d/2$,

$$\frac{1}{2}d = \frac{1}{2}d \cos(0 + \phi) \Rightarrow \cos(\phi) = 1$$

So, the phase is zero, that is $\phi = 0$.

So, $x = \frac{1}{2} d \cos(\omega t)$

To find the time, we have to substitute with x into (1) and solve for t as,

$$\frac{1}{4}d = \frac{1}{2}d \cos(\omega t)$$

$$\Rightarrow \cos(\omega t) = \frac{1}{2}$$

$$\Rightarrow \omega t = \cos^{-1}\left(\frac{1}{2}\right)$$

$$\Rightarrow t = \frac{1}{\omega} \cos^{-1}\left(\frac{1}{2}\right)$$

$$\Rightarrow t = \frac{1}{0.503 \text{ rad/h}} \cos^{-1}\left(\frac{1}{2}\right)$$

$$\therefore t = 2.082 h$$

(Ans:)

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Problem 8: Two particles oscillates in simple harmonic motion along a common straight-line segment of length A. Each particle has a period of 1.5 s, but they differ in phase by $\pi/6$ rad. (a) How far apart are they (in terms of A) 0.50 s after the lagging particle leaves one end of of the path? (b) Are they then moving in the same direction toward each other, or away from each other?

Answer: The amplitude of the oscillations for each of the particles is A/2, each particle has a period of $T = 1.5$ s but they differ in phase by $\pi/6$ rad. For each particle the coordinates,

$$x_1 = \frac{A}{2}\cos(\omega t)$$

$$x_2 = \frac{A}{2}\cos\left(\omega t + \frac{\pi}{6}\right)$$

Now,

$$x_1 = \frac{A}{2}\cos\left(\frac{2\pi t}{T}\right)$$

$$x_2 = \frac{A}{2}\cos\left(\frac{2\pi t}{T} + \frac{\pi}{6}\right)$$

We know, $\omega = \frac{2\pi}{T}$

(a) The coordinates of the particles at $= 0.50$ s are,

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$$x_1 = \frac{A}{2}\cos\left(\frac{2\pi \times 0.50s}{1.5s}\right) = -0.25A$$

$$x_2 = \frac{A}{2}\cos\left(\frac{2\pi \times 0.50s}{1.5s} + \frac{\pi}{6}\right) = -0.43A$$

their separation at that is, therefore,

$$\Delta x = x_1 - x_2$$

$$\Rightarrow \Delta x = -0.25A + 0.43A$$

$$\therefore \Delta x = 0.18A$$

(Ans:)

(b) at $t = 0.50s$, the derivative of the position at this time,

$$v_1 = \frac{dx_1}{dt} = -\frac{\pi A}{T}\sin\left(\frac{2\pi t}{T}\right)$$

$$\Rightarrow v_1 = -\frac{\pi A}{T}\sin\left(\frac{2\pi \times 0.50s}{1.5s}\right)$$

$$\Rightarrow v_1 = \frac{-0.866\pi A}{T}$$

and,

$$v_2 = \frac{dx_2}{dt}$$

$$\Rightarrow v_2 = -\frac{\pi A}{T}\sin\left(\frac{2\pi t}{T} + \frac{\pi}{6}\right)$$

$$\Rightarrow v_2 = -\frac{\pi A}{T}\sin\left(\frac{2\pi \times 0.50s}{1.5s} + \pi/6\right)$$

$$\therefore v_2 = -\frac{0.5\pi A}{T}$$

As both v_1 and v_2 are negative

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So the particles are moving in the same direction.

Problem 9: Two particles execute simple harmonic motion of the same amplitude and frequency along close parallel lines. They pass each other moving in opposite directions each time their displacement is half their amplitude. What is their phase difference?

Answer: The two particles' displacement is half their amplitude, that is,

$$x_1 = x_2 = \frac{1}{2} x_m \quad \text{——— (1)}$$

and the coordinates of the particles are,

$$x_1 = x_m \cos(\omega t + \phi_1)$$
$$x_2 = x_m \cos(\omega t + \phi_2)$$

$\therefore \cos(\omega t + \phi_1) = \cos(\omega t + \phi_2) = \frac{1}{2}$ [Applying eq (1)]

And the velocities of the particles are given by,

$$v_1 = -x_m \omega \sin(\omega t + \phi_1)$$
$$v_2 = -x_m \omega \sin(\omega t + \phi_2)$$

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it is given that at the instant when pass each others, we have $v_1 - v_2 \neq 0$

$$\therefore \sin(\omega t + \phi_1) = -\sin(\omega t + \phi_2)$$

Now, we see that one phase is $+\pi/3$ and another phase is $-\pi/3$, taking the phase difference we get,

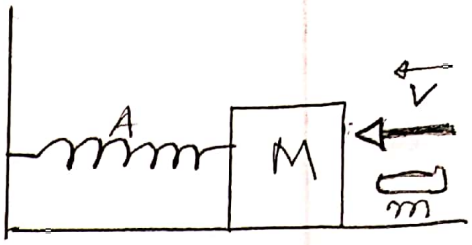
$$\Delta\phi = \frac{\pi}{3} - \left(\frac{-\pi}{3}\right)$$

$$\therefore \Delta\phi = \frac{2\pi}{3} \quad \text{(Ans:)}$$

Problem 10: A block of mass $M = 5.4$ kg, at rest on a horizontal frictionless table, is attached to a rigid support by a spring of constant $K = 6000$ N/m. A bullet of mass $m = 9.5$ g and velocity of magnitude 630 m/s strikes and embedded in the block. Assuming the compression of the spring is negligible until the bullet is embedded, determine (a) the speed of the block immediately after the collision and (b) the amplitude of the resulting simple harmonic motion.

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Answer: From the conservation of momentum we know,



$$P_i = P_f$$

(a) $\Rightarrow m v_i = (M+m) V$ [v_i is the intial speed of the bullet]

$\Rightarrow V = \left(\frac{m v_i}{M+m}\right)$

$\Rightarrow V = \left(\frac{9.5 \times 630}{5.4 \times 10^3 + 9.5}\right)$

Given,
$m = 9.5$ g
$M = 5.4$ kg
$v_i = 630$ m/s

$\therefore V = 1.106$

$\simeq 1.1$ m/s (Ans:)

(b) Also from the conservation of energy we get,

$$\frac{1}{2}(M+m)V^2 = \frac{1}{2}KA^2$$

$$\Rightarrow \frac{1}{2} \times (m+M) \frac{m v^2}{(m+M)^2} = \frac{1}{2} \times KA^2$$

$$\Rightarrow A = \frac{mv}{\sqrt{K(m+M)}} \quad \because A^2$$

$$= \frac{(9.5 \times 10^{-3} \text{Kg})(630 \text{ m/s})}{\sqrt{(6000 \text{ N/m})(9.5 \times 10^{-3} \text{Kg} + 5.4 \text{kg})}}$$

$$= 3.3 \times 10^{-2} \text{ m} \quad \text{(Ans:)}$$

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